Fiscal Year:	FY 2020	**Task Last Updated:**	FY 02/18/2020
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	10/15/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/15/2019 per PI (Ed., 1/28/19) NOTE: end date change per P. Downey/MSFC (Ed., 1/12/2015) NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin Ph.D. (University of Alabama Huntsville) Cröll, Arne Ph.D. (University of Freiburg, Germany) Sorgenfrei, Tina Ph.D. (University of Freiburg, Germany)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10%. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms that lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Description:	EXECUTIVE SUMMARY Reduced Defects in Germanium-Silicon (RDGS) was the original name of the project with Dr. Frank Szofran and later Dr. Martin Volz. The name was changed to ICESAGE (Influence of Containment on the Growth of Silicon-Germanium) and is a materials science flight investigation concerning the microgravity processing of Ge1-xSix alloys by directional solidification methods. It was sponsored by NASA's Space Life and Physical Sciences Research and Applications Division. The flight experiments were to be conducted in the Low Gradient Furnace (LGF) in the Materials Science Research Rack (MSRR) onboard the International Space Station (ISS). It was an international collaborative project from the start, with an equivalent project at the European Space Agency (ESA), also named RDGS. The coupled projects involved 23 investigators from 9 organizations in 4 countries over the years. The investigation continues to be known as RDGS by the European Space Agency (ESA) and the participating European national space agencies. The lead organization in Europe is the University of Freiburg, Germany. The Requirements Definition Review (RDR) was held on September 26, 2012. The panel report did not identify any science or engineering feasibility issues and it was recommended to proceed on to an integrated design review (IDR). After the RDR, extensive laboratory experiments and thermal modelling were conducted to develop and optimize the design of the flight ampoules. By 2017, these designs were complete and flight prototype ampoules had been built and tested in the laboratory. Thermal modeling by the engineering team demonstrated the feasibility of conducting the proposed crystal growth processes in LGF in the MSRR. The accomplishments of the ICESAGE/RDGS team are extensive. To date, experimental and theoretical studies have resulted in 70 refereed journal publications, 3 book chapters, and 10 conference proceedings, published between 2001 and 2019. These are listed in the Cumulative Bibliography for RDGS with Principal Investigator Martin Volz, with PI Frank Szofran in the 2004 Task Book record (https://) , and the RDGS Task Book Archives record with PI Dr. Szofran (Ed. Note: contact taskbook@nasaprs.com for access to the Archives). In 2017 ICESAGE was cancelled as a NASA flight investigation although resources were continued for the closeout of U.S. participation. Note that ICESAGE/RDGS continues as a European microgravity science flight investigation. HISTORY OF THE INVESTIGATION This investigation originated as an international collaboration led by the Marshall Space Flight Center (MSFC) and the University of Freiburg, Germany. On the U.S. side, the proposal "Influence of Containment on Defects in GeSi Crystals: Comparison of Detached Bridgman and Floating-Zone Growth" was selected from NASA NRA-96-HEDS-02 and was initiated in 1998. On the German side, a proposal with the same title was selected from the ESA 1998 Announcement of

Task Progress:	Opportunity (AO) and was registered under the reference number AO-99-034. The majority of team members of the ESA project were also a member of the NASA project and vice versa. Since the selection of the projects, the participation of many individuals and several organizations has changed over the years. The NASA Science Concept Review (SCR) for this investigation was held December 19-20, 2000. The SCR report stated that "the panel were very impressed by the high profile program," and recommended continuation of the project to the Requirements Definition Review (RDR). Cancellation of the NASA Microgravity Materials Science Program in 2004 resulted in the termination of this project; PI at the time was Frank Szofran. Funding for this investigation was re-initiated in 2008 with PI Martin Volz. In the original research plan, the development of the Bridgman and detached Bridgman experiments for the ISS (in the LGF insert of MSL) was to be done at MSFC, and the development of the ISS float-zone (FZ) experiments (in the FMF insert for MSL) together with the use of external fields, was to be done in Europe. This was not a clear cut separation, as Bridgman and vertical gradient freeze (VGF) experiments have been done in Germany at both the University of Freiburg and the University of Freiberg, including microgravity precursor experiments on Russian unmanned capsules (FOTON). Ground-based FZ experiments have been done at the University of Freiburg and microgravity FZ experiments have been conducted as part of the MAXUS series of suborbital rocket flights. A comparison of the results of detached Bridgman, normal Bridgman, and FZ growth methods is a primary objective of this investigation. The RDR was held on September 26, 2012. There was both a science panel and an engineering panel for the review. The science panel made the following determination. "The project is mature and appropriate to enter into the flight development phase. The project is scientifically sound, and the space experiments will provide data useful to both scientific community and industry. We are not aware of any science or engineering feasibility issues. The proposed engineering plan is solid." The engineering panel stated that "the engineering panel believes that the proposed experiment design is feasible and will meet the experiment objectives." After the RDR, extensive laboratory experiments were conducted to develop and optimize the design of the flight ampoules. By 2017, these designs were complete and flight prototype ampoules had been built and tested in the laboratory. Thermal modeling by the engineering team demonstrated the feasibility of conducting the proposed crystal growth processes in LGF in the MSRR. In September 2017 the ICESAGE/RGDS team was informed that ICESAGE was cancelled as a NASA flight investigation. Resources were to be provided for the closeout of U.S. involvement in the investigation but microgravity flight experiments would not be supported. Note that ICESAGE/RDGS continues as a European microgravity science flight investigation. [Ed. note: compiled from final report dated March 2020]

Bibliography Type:	Description: (Last Updated: 07/09/2021)
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